

Ziming Liu (刘子铭)

E-mail: e0732706@u.nus.edu

Homepage: <https://maruyamaaya.github.io/>

Google Scholar: <https://scholar.google.com/citations?user=UPFptUwAAAAJ>

Twitter/X: @lzm_mlsys

National University of Singapore / Peking University

Education

National University of Singapore, School of Computing

➤ Ph.D. in Computer Science Jan. 2023 – Present

National University of Singapore, School of Computing

➤ Master's degree in computer science (Artificial Intelligence) Aug. 2021 – Jan. 2023

Peking University, School of Electronics Engineering and Computer Science

➤ B.S. in Computer Science and Technology Sep. 2016 – Jul. 2020

Industry Experience

Tencent

May. 2026 – Current

Research Intern, Hunyuan – AI Infra Group

Shanghai Qijizhifeng Co.

Jan. 2025 – Nov. 2025

Research Intern, Infra Group

Microsoft Research Asia.

May. 2024 – Nov. 2024

Research Intern, System Group

Rewarded “Star of Tomorrow” certificate (Top 10% intern)

HPC-AI Tech.

May. 2022 – Dec. 2022

Research Intern, Infra Group

ByteDance Inc.

Aug. 2020 – Jul. 2021

Machine Learning Engineer, Lark

Research Interests

Machine Learning System and High Performance Computing.

Including distributed model training (parallelism schemes) / inference and serving systems.

Award

ByteDance Scholarship Award 2025 (“字节奖学金” 2025, 20 awardees in total)

NUS Research Achievement Award 2025

MSRA Star of Tomorrow Award 2024

NUS SoC Teaching Fellowship 2024

Highlight Research Experience

Expert-as-a-Service:

Towards Efficient, Scalable, and Robust Large-scale MoE Serving

May. 2025 – Dec. 2025

First Author.

Objective: Efficient, scalable, and robust MoE Serving System.

- Our system disaggregates MoE modules into independent, stateless services. This design enables fine-grained resource scaling and provides inherent fault tolerance by decoupling compute units.
- The architecture is powered by a high-performance, CPU-free peer-to-peer communication library that ensures minimal overhead and high throughput.
- EaaS incurs less than a 2% throughput reduction under simulated hardware failures that would otherwise halt monolithic architectures. It further saves up to 37.5% of computing resources through dynamic fine-grained adaptation to serving traffic, demonstrating strong resilience for large-scale MoE deployment in production.

StarTrail:

Concentric Ring Sequence Parallelism for Efficient Near-Infinite-Context Transformer Model Training

First Author. Advisor: Presidential Young Prof. You Yang, Prof. James Demmel

Dec. 2023 – June.2024

Objective: We develop a multi-dimensional sequence parallel system to reduce the communication volume and improve overall efficiency for long-sequence Transformer model training.

- This paper has been accepted by NeurIPS '25.
- We conceptualize Attention computation as a novel instance of the traditional n-body problem, providing fresh insights into optimizing and parallelizing Attention computation.
- We introduce a near-infinite-context training system for Transformer models, featuring a groundbreaking multi-ring sequence parallelism scheme.
- Preliminary results indicate that our StarTrail system outperforms Ring Attention by up to 77.12%, showcasing its efficacy and scalability.

Hanayo: Harnessing Wave-like Pipeline Parallelism for Enhanced Large Model Training Efficiency

First Author. Advisor: Presidential Young Prof. You Yang

Dec. 2022 – Apr.2023

Objective: We develop a new pipeline parallel technique to solve the problem the bubbles in existing pipeline model training techniques and achieve SOTA results in multiple tasks.

- This paper has been accepted by SC '23(The International Conference for High Performance Computing, Networking, Storage, and Analysis).
- We introduce a wave-like pipeline scheme that achieves a low bubble ratio and high performance in large model training.
- Utilizing the action list, Hanayo's runtime system can support nearly all pipeline parallel algorithms while optimizing performance through features such as asynchronous communications.
- Experimental results demonstrate that Hanayo achieves up to a 30.4% performance improvement over the current state-of-the-art pipeline parallelism implementation.

WeiPipe: Weight Pipeline Parallelism for Communication-Effective Long-Context Large Model Training

Co-first Author. Advisor: Presidential Young Prof. You Yang and Prof. Rong Zhao

Apr. 2024 – Nov.2024

Objective: We introduce weight-pipeline parallelism (WeiPipe) that transitions from an activation-passing pipeline to a weight-passing pipeline in long-context scenarios to reduce the communication volume and enhance efficiency.

- This paper has been accepted by PPoPP '25(ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming).
- We propose WeiPipe, WeiPipe-Interleave, and WeiPipe-Zero-Bubble that reduces the bubble ratio and relieve the

communication requirements

- Experimental results demonstrate that WeiPipe can improve training efficiency by about 30%-80% compared to state-of-the-art PP and maintain weak and strong scalability under long-context scenarios.

Region-Adaptive Sampling for Diffusion Transformers

First Author. Advisor: Dr. Yuqing Yang

May. 2024 – Dec. 2024

Objective: Efficient Diffusion Transformer Inference.

- This paper has been accepted by CVPR '26.
- we introduce RAS, a novel, training-free sampling strategy that dynamically assigns different sampling ratios to regions within an image based on the focus of the DiT model.
- We evaluate RAS on Stable Diffusion 3 and Lumina-NextT2I, achieving speedups up to 2.36x and 2.51x, respectively, with minimal degradation in generation quality.

Publication

Hanayo: Harnessing Wave-like Pipeline Parallelism for Enhanced Large Model Training Efficiency

Ziming Liu*, Shenggan Cheng*, Haotian Zhou, and Yang You

SC '23, *In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 2023

*: Equal Contribution.

Cited: 71

StarTrail: Concentric Ring Sequence Parallelism for Efficient Near-Infinite-Context Transformer Model Training

Ziming Liu, Shaoyu Wang, Shenggan Cheng, Zhongkai Zhao, Yang Bai, Xuanlei Zhao, James Demmel, Yang You

NeurIPS '25, Arxiv: 2407.00611, 2024

Cited: 3

Region-Adaptive Sampling for Diffusion Transformers

Ziming Liu, Yifan Yang, Chengruidong Zhang, Yiqi Zhang, Lili Qiu, Yang You, Yuqing Yang

CVPR '26, Arxiv:2502.10389, 2024

Cited: 11

WeiPipe: Weight Pipeline Parallelism for Communication-Effective Long-Context Large Model Training

Junfeng Lin*, **Ziming Liu***, Yang You, Jun Wang, Weihao Zhang, Rong Zhao

PPoPP '25, *ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming*

*: Equal Contribution.

Cited: 11

Concerto: Automatic Communication Optimization and Scheduling for Large-Scale Deep Learning

Shenggan Cheng, Shengjie Lin, Lansong Diao, Hao Wu, Siyu Wang, Chang Si, **Ziming Liu**, Xuanlei Zhao, Jiangsu Du, Wei Lin, Yang You

ASPLOS 2025, *ACM International Conference on Architectural Support for Programming Languages and Operating Systems*

Cited: 8

HelixPipe: Efficient Distributed Training of Long Sequence Transformers with Attention Parallel Pipeline Parallelism

Geng Zhang, Shenggan Cheng, Xuanlei Zhao, **Ziming Liu**, Yang You

PPoPP '26, *ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming*

Cited:1

Respec: Towards optimizing speculative decoding in reinforcement learning systems

Qiaoling Chen, Zijun Liu, Peng Sun, Shenggui Li, Guoteng Wang, **Ziming Liu**, Yonggang Wen, Siyuan Feng, Tianwei Zhang

MLSys 2026, *In Proceedings of Machine Learning and Systems 2026*

Cited: 2

HeteGen: Efficient Heterogeneous Parallel Inference for Large Language Models on Resource-Constrained Devices

Xuanlei Zhao, Bin Jia, Haotian Zhou, **Ziming Liu**, Shenggan Cheng, and Yang You

MLSys 2024, *In Proceedings of Machine Learning and Systems 2024*

Cited: 46

DSP: Dynamic Sequence Parallelism for Multi-Dimensional Transformers

Xuanlei Zhao, Shenggan Cheng, Zangwei Zheng, Zheming Yang, **Ziming Liu**, and Yang You
2024

ICML 2025, Arxiv: 2403.10266, 2024

Cited: 14

Preprints

Expert-as-a-service: Towards efficient, scalable, and robust large-scale moe serving

Ziming Liu, Boyu Tian, Guoteng Wang, Zhen Jiang, Peng Sun, Zhenhua Han, Tian Tang, Xiaohe Hu, Yanmin Jia, Yan Zhang, He Liu, Mingjun Zhang, Yiqi Zhang, Qiaoling Chen, Shenggan Cheng, Mingyu Gao, Yang You, Siyuan Feng

Arxiv: 2509.17863, 2025

Cited: 4

EnergonAI: An Inference System for 10-100 Billion Parameter Transformer Models

Jiangsu Du, **Ziming Liu**, Jiarui Fang, Shenggui Li, and Yongbin Li, Yutong Lu, Yang You

Arxiv: 2301.08658 , 2022

Cited: 9

ATP: Adaptive Tensor Parallelism for Foundation Models

Shenggan Cheng, **Ziming Liu**, Jiangsu Du, and Yang You

Arxiv: 2209.02341, 2023

Cited: 17